Section B
Subject: Science

- **Answer from the following questions in a separate sheet and attach them to the worksheet.**

1. Choose the correct answer.

A. Which one of the following condition is not essential to grow maize?
a. High temperature b. Humidity c. Rainfall d. Low temperature

B. If light falls perpendicularly on a plane mirror, what will be the angle at which it will be reflected?
a. 45º b. 180º c. 0º d. 360º

C. A light ray strikes a reflective plane surface at an angle of 32 with the surface. Calculate the angle of incidence.
a. 58º b. 36º c. 22º d. 32º

D. The term used for the process of separation of grains from chaff is _____
a. sieving b. winnowing c. threshing d. hand picking

E. How many images will be formed when two plane mirrors are placed parallel to each other?
a. 2 b. 10 c. 16 d. Infinite

F. Purest form of carbon is _____.
a. coal b. charcoal c. coke d. petrol

i. Refining of petroleum is done in a_____ column.
a. fractionating b. distillation c. separating d. Reserving

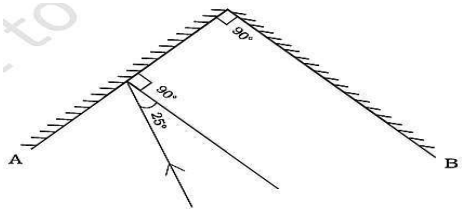
H. Which of the following statement is not true for organic manure?
a. It enhances water holding capacity of soil. b. It has a balance of all plant nutrients.
c. It provides humus to soil. d. It improves texture of soil.

2. Fill in the blanks in the following.

A. A person 1 m in front of a plane mirror seems to be _____ m away from his image.
B. The tool used with a tractor for sowing seeds in a field _____.
C. The size of the pupil becomes _____ when you see in dim light.
D. Night birds have _____ cones than rods in their eyes.
E. The process of loosening and turning of soil is called_____.
F. Process of separation of different constituents from petroleum is called _____ .
G. Least polluting fuel for vehicle is _____ .

3. Answer the following questions.

A. What is the most comfortable distance at which one can read with normal eye?
B. What is the unit of luminous intensity?
C. What are stubs? Why is burning of stubs on field harmful?
D. What is normal?
E. Why is it important to sow seeds at the correct distance/ depth?
F. Name the petroleum product used as fuel for stoves, lamps and jet aircrafts.
G. Name the products obtained and their uses when coal is processed in industry.
H. What steps would you suggest for the judicious use of fossil fuels?
I. Why petrol is exhaustible natural resource, whereas sunlight is not? Explain.
J. What is accommodation of eye?
K. What is the role of ciliary muscles in human eye?
L. What is the purpose of storage of grains? How are grains stored? What is the function of adding dried neem leaves to store grains?
M. Which activity of farmers can promote growth of earthworms and microorganisms in the field?
N. How many images are formed when two plane mirrors are placed at 45º?
O. What is meant by tilling? How is this done?
P. List any two factors on which time and frequency of irrigation depends.
Q. Two mirrors A and B are placed at right angles to each other as shown in figure. A ray of light incident on mirror A at an angle of 25º falls on mirror B after reflection. What would be the angle of reflection for the ray reflected from mirror B?



4. In the following, question two statements are given- one labelled Assertion (A) and the other labelled Reason (R). Choose the correct answer for these questions from options given below.

a. A is correct and R is the correct explanation for A.
b. A is correct but R is not the correct explanation for A.
c. A is correct R is wrong.
d. A is wrong R is correct.

A. Assertion (A): Rainbow forms in the sky only after rains.
Reason (R): Water drops suspended in the air split sun rays into seven colours.

B. Assertion (A): Many useful substances are obtained from petroleum and natural gas called as Petrochemicals.
Reason (R): These are used in the manufacture of detergents, fibres, polythene and other man-made plastics.
C. Assertion (A): Manure is considered a better option than chemical fertilizers for enriching soil. Reason (R): Manure improves the texture and water retention capacity of the soil.

5. Answer the following questions on the basis of your understanding of the following paragraph and the related studied concepts.

A. Two friends Rahul and Rohan are going to school. In the morning, traffic is slow but a van is moving at a high speed and everyone is giving space to it responsibly without any delay. Rahul feels surprised to see this. He tries, but is unable to read what is written on it. Rohan then explained the word 'ambulance' is written in reverse so that it can be read clearly in the rear-view mirror.

a. Why is the word ambulance written in reverse?
b. Why is everyone on road giving space to ambulance?
c. What the nature of image is formed by plane mirror?
d. Write four characteristics of image formed by plane mirror.

B. Akash went for a walk in fields in his village. There he saw a farmer spraying something in his fields. Akash asked the farmer about it. The farmer told him that there were several undesirable plants (weeds) in his fields, so he was spraying weedicides. Akash noticed that the farmer hadn't covered his mouth and nose. Akash had studied in his book that weedicides are chemicals which are harmful, so he suggested the farmer to cover his mouth and nose during spraying.

a. Which other method can be used by farmer to remove weeds?
b. Why did Akash suggest the farmer to cover his nose and mouth while spraying weedicides?
c. Akash's mother uses vegetables after cleaning well with water. Why does she do so?
d. Name one chemical used as weedicide.

C. Radha prefers to go to school by pooling car. On other occasions she likes to opt for public transport rather than private transport. For short distances, he walks or uses bicycle.

a. What are the benefits of carpooling?
b. Which mean of transport do you think is best for short distances?
c. How on our end we can reduce the consumption of fossil fuels?
d. What values of Radha are shown here?